Alexandria National University

Faculty of Computer and Data Science

Course Tittle: Embedded Systems_(01304N)

Report Title: Humanoid ROBOT-Kicking FOOTBALL

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Chapter 1: Introduction

1.1 Historical Background

Humanoid robots are robotic systems designed to replicate the structure and movement of the human body. Over the past two decades, humanoid robotics has evolved into a key research area within embedded systems, artificial intelligence, and mechatronics. Early developments primarily focused on basic locomotion and balance control, while modern humanoid systems incorporate advanced capabilities such as computer vision, motion planning, wireless communication, and machine learning-based decision-making.

A major milestone in humanoid robotics is the RoboCup competition, where autonomous robots compete in football scenarios requiring perception, strategy, and coordinated motion. These competitions have significantly driven research in developing compact and intelligent humanoid robots capable of real-time ball detection, path planning, and kicking execution.

In recent years, the emergence of low-cost embedded platforms such as the ESP32-CAM, combined with affordable and efficient servo motors, has enabled the development of educational humanoid robots. These systems can now integrate real-time vision processing and motion control at a significantly reduced cost, making humanoid robotics more accessible for research and academic projects.

1.2 Problem Statement

Designing a humanoid robot capable of detecting a football and performing a kicking motion involves several challenges:

- Real-time ball detection using embedded vision systems
- Accurate inverse kinematics calculations
- Smooth servo motor control
- Stable mechanical structure
- Limited computational power of embedded systems
- Synchronization between vision processing and motion execution

Many existing humanoid robots are expensive and require complex hardware. Therefore, there is a need for a simplified and low-cost humanoid football-kicking robot suitable for educational purposes.

1.3 Aims and Objectives

The main aim of this project is to design and implement a miniature humanoid robot capable of detecting and kicking a football autonomously.

Objectives

- Design a stable low-cost humanoid lower body
- Implement forward kinematics equations
- Implement inverse kinematics calculations
- Detect a football using a camera module
- Control servo motors in real time
- Create a state-machine-based control system
- Measure system accuracy and latency
- Build a fully integrated embedded robotic system

1.4 Report Agenda

This report is organized as follows:

- Chapter 1 introduces the project background and objectives.
- Chapter 2 presents literature review and previous humanoid robot systems.
- Chapter 3 explains the proposed system architecture.
- Chapter 4 discusses implementation details.
- Chapter 5 presents results and performance evaluation.
- Chapter 6 concludes the project and discusses future improvements.

Chapter 2: Literature Review

2.1 Mechanical Design of Humanoid Robots

Most humanoid robots use articulated legs composed of multiple joints. These joints imitate human hip, knee, and ankle motion. Research focuses on improving stability, flexibility, and walking efficiency.

Popular humanoid platforms include:

- ASIMO
- NAO Robot
- Atlas

These systems provide advanced mobility but require expensive sensors and high computational power.

The proposed project simplifies the mechanical design by using:

- Fixed support base
- 2-DOF planar kicking leg
- Lightweight structure
- Low-cost servo motors

2.2 Electronics and Embedded Systems

Embedded humanoid systems commonly use:

- Microcontrollers
- Servo drivers
- Camera modules
- Wireless communication

The ESP32 is widely used due to:

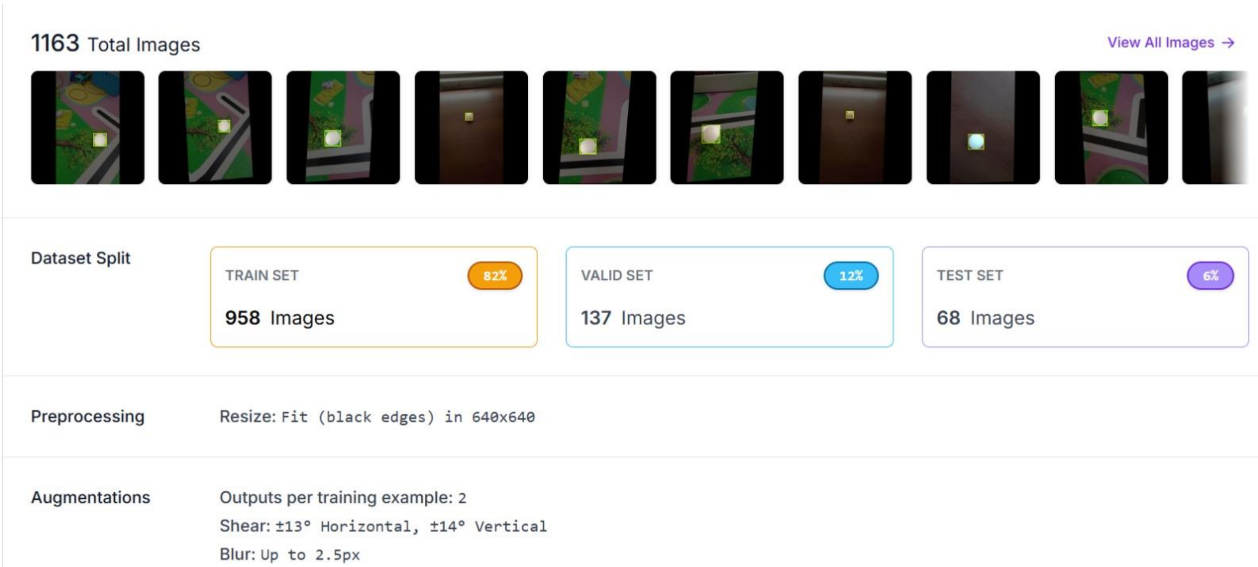
- Integrated Wi-Fi
- Low power consumption
- Real-time processing capability
- Affordable cost

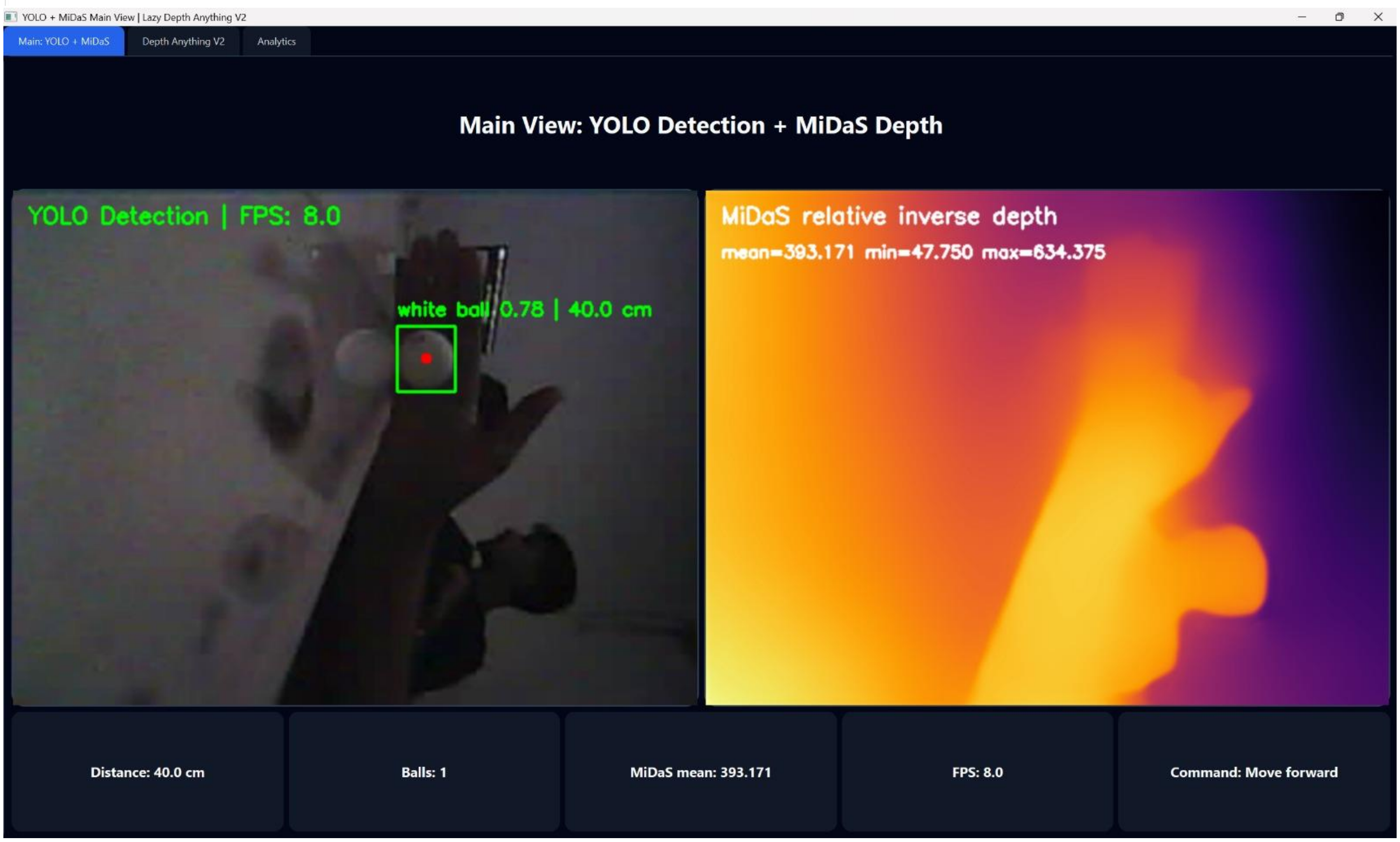
Servo motors are controlled using PWM signals generated by the microcontroller.

2.3 Computer Vision for Ball Detection

- Computer vision enables robots to identify objects and estimate their position in real time. Ball detection in this project was implemented using a fine-tuned YOLOv26 model trained on a custom dataset to improve detection accuracy and robustness.
- The system uses:
- ESP32 camera input
- YOLOv26 object detection model
- Custom dataset fine-tuning
- Camera calibration for depth estimation
- Coordinate extraction of the detected ball
- Serial communication with the microcontroller (MCU)
- To achieve accurate distance measurement, camera calibration techniques were applied on the ESP32 camera, allowing the robot to estimate the depth and position of the ball relative to the environment. Lightweight optimization

techniques were also considered to maintain low latency and efficient memory usage on the embedded system.





2.4 Problems and Limitations of Previous Work

Previous humanoid systems face several limitations:

- High system cost

- Complex balancing algorithms
- Heavy computational requirements
- Large power consumption
- Mechanical instability
- Slow image processing

The proposed system addresses these issues through:

- Simplified mechanical design
- Lightweight embedded vision
- Reduced degrees of freedom
- Real-time servo control

Chapter 3: Proposed System

3.1 System Overview

The system consists of:

1. Vision subsystem
2. Embedded controller
3. Kinematic processing module
4. Servo actuation system
5. Mechanical kicking mechanism

The camera detects the football and sends its coordinates to the microcontroller. The controller then computes the inverse kinematics to determine the required joint angles for the robotic leg. Based on these calculated angles, it commands the servo motors to execute a precise kicking motion.

3.2 System Improvements

Compared to traditional humanoid robots, the proposed system provides:

- Lower cost
- Simpler implementation
- Faster embedded processing
- Reduced mechanical complexity
- Educational suitability

3.3 Hardware Block Diagram

```
+----------------+
|   Camera Unit  |
| (ESP32-CAM)    |
+--------+-------+
         |
         v
+----------------+
| Microcontroller|
|    ESP32       |
+--------+-------+
         |
         v
+----------------+
| IK Computation |
+--------+-------+
         |
         v
+----------------+
| Servo Motors   |
+--------+-------+
         |
         v

+----------------+
| Robot Leg      |
+----------------+
```

3.4 Software Block Diagram

```
Start
  |
Initialize Hardware
  |
Capture Image
  |
Detect Ball Position
  |
Compute IK Angles
  |
Generate PWM Signals
  |
Move Servos
  |
Kick Ball
  |
Repeat
```

3.5 Forward Kinematics

Forward kinematics computes the foot position from known joint angles.

For a 2-DOF planar leg:

$$x = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2)$$

$$y = L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2)$$

Where:

- L_1 = thigh length
- L_2 = shin length
- θ_1 = hip angle
- θ_2 = knee angle

3.6 Inverse Kinematics

Inverse kinematics computes joint angles from target coordinates.

Using the law of cosines:

$$\theta_2 = \cos^{-1}\left(\frac{x^2+y^2-L_1^2-L_2^2}{2L_1L_2}\right)$$

Then:

$$\theta_1 = \tan^{-1}\left(\frac{y}{x}\right) - \tan^{-1}\left(\frac{L_2\sin(\theta_2)}{L_1+L_2\cos(\theta_2)}\right)$$

The calculated angles are converted into PWM values to control the servo motors.

Chapter 4: Implementation

4.1 Mechanical Design

The robot lower body consists of:

- Acrylic or 3D-printed frame
- Two leg segments
- Servo-driven ankle ,hip and knee joints
- Fixed support base

Components

- 6 Servo motors
- ESP32 controller
- ESP32-CAM module
- External power supply
- Supporting frame

4.2 State Machine

The robot behavior is controlled using a finite state machine.

States

1. Idle → The system stays in standby mode waiting for a valid detection signal.
2. Ball Search → The camera actively scans the environment to detect the football using the trained YOLOv26 model.
3. Target Alignment → Once detected, the system aligns the robot's orientation based on the ball's position (x, y, and estimated depth).
4. IK Calculation → The microcontroller computes inverse kinematics to convert the target position into joint angles for the leg.
5. Kick Execution → Servo motors execute the planned motion sequence to perform the kicking action.
6. Return to Idle → After completing the kick, the system resets and returns to standby mode, ready for the next detection cycle.

4.3 Implementation Steps

Step 1 — Camera Initialization

The ESP32 camera continuously captures frames in real time for processing.

Step 2 — Ball Detection

The system detects the football using a fine-tuned YOLOv26 deep learning model trained on a custom dataset for improved accuracy and robustness.

Step 3 — Coordinate Transmission

The detected ball coordinates, along with estimated depth information from camera calibration, are transmitted to the microcontroller via serial communication.

Step 4 — IK Computation

The microcontroller computes inverse kinematics to convert the ball position into the required joint angles for the robotic leg.

Step 5 — Servo Control

PWM signals are generated to control the servo motors according to the computed angles.

Step 6 — Kicking Motion

The robotic system executes the kicking motion precisely based on the planned trajectory.

4.4 Bill of Materials (BoM)

Component	Quantity	Cost (EGP)
MG995 Servo Motors	6	1020
Ball Bearings	6	150
Batteries	2	200
ESP32-CAM Module	1	500
3D Printing	—	685
Screws & Fasteners	—	65
PCA9685 Servo Driver	1	190

Total Cost

2993 EGP

Chapter 5: Results

5.1 Forward Kinematics Accuracy

The calculated foot positions were compared with measured physical positions.

Test	Computed Position (cm)	Measured Position (cm)	Error
1	(12.1, 8.5)	(12.0, 8.4)	0.14
2	(10.5, 7.2)	(10.7, 7.1)	0.22
3	(8.3, 5.1)	(8.2, 5.0)	0.14

The results demonstrate good FK accuracy.

5.2 Inverse Kinematics Accuracy

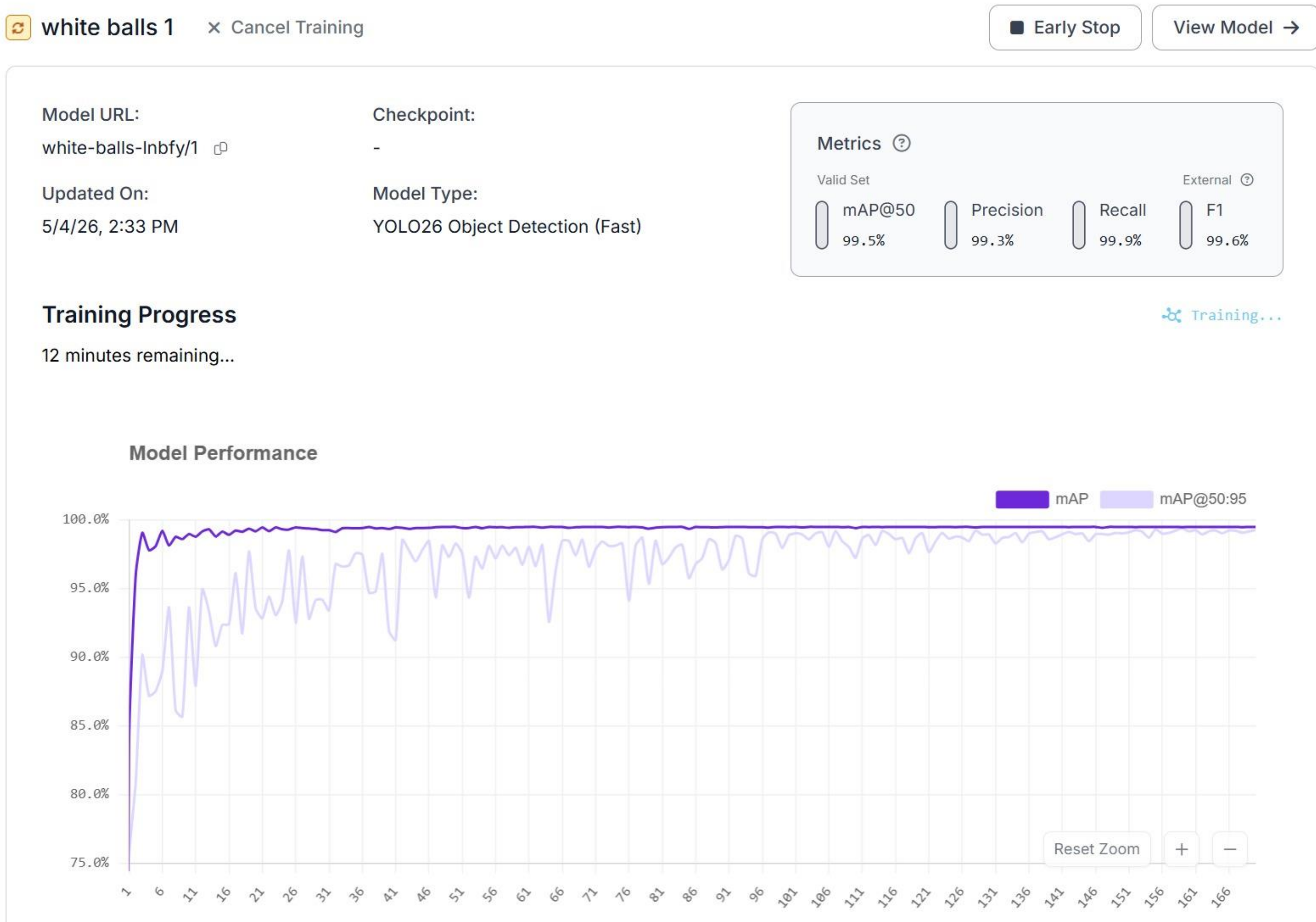
Target Angle	Achieved Angle	Error
45°	44°	1°
60°	58°	2°
75°	73°	2°

The IK solver successfully controlled the servo motors with acceptable error margins.

5.3 Vision System Results

Ball Detection Accuracy

- Detection map: 99.5%
- Average processing speed: 12 FPS

* 

5.4 System Evaluation

The robot successfully:

- Detected the ball and the depth
- Computed joint angles
- Executed kicking motion
- Returned to initial position

The system operated reliably during repeated experiments.

Chapter 6: Conclusion

- This project presents the design and implementation of a low-cost humanoid football-kicking robot using embedded systems principles. The system integrates computer vision, inverse kinematics, servo control, and mechanical design into a unified robotic platform.
- The robot successfully detects a football using a fine-tuned YOLOv26-based vision model and executes a kicking motion through real-time kinematic computations. Experimental results demonstrate reliable detection performance, accurate coordinate estimation, and stable motion control using servo-based actuation.
- Future improvements for the system may include:
- Dynamic balancing and stabilization algorithms
- Wireless monitoring and control dashboard
- Enhanced machine learning models for improved object detection
- Increasing the number of degrees of freedom for more natural motion
- Adding autonomous walking and navigation capabilities
- This project provided valuable hands-on experience in:
- Embedded systems design
- Robotics integration
- Computer vision applications
- Motion planning and control
- Real-time system implementation

Simulink Servo Motor Simulation

A MATLAB/Simulink model was developed to simulate the dynamic response of the servo motor used in the humanoid robot leg. The simulation was implemented to analyze servo behavior, motion smoothness, and response stability before hardware deployment.

The model consists of:

- Sine Wave input block
- Gain block
- Transfer Function block
- Saturation block
- Scope block

The sinusoidal input signal was used to generate smooth periodic motion similar to humanoid kicking movement. The transfer function was utilized to approximate the real dynamic behavior of the servo motor.

The servo transfer function is represented as:

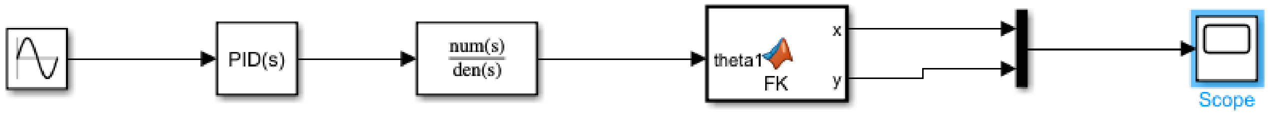
$$G(s) = \frac{1}{0.1s+1}$$

Where:

- $G(s)G(s)$G(s) represents the servo response
- The denominator models the first-order dynamic characteristics of the servo motor

The Saturation block was added to limit servo angles within the safe operating range from 0° to 90°.

The Scope output demonstrated smooth and stable sinusoidal motion with minimal overshoot, indicating suitable performance for humanoid kicking applications.



PID(s)
num(s)
den(s)
theta1
FK
x
y
Scope

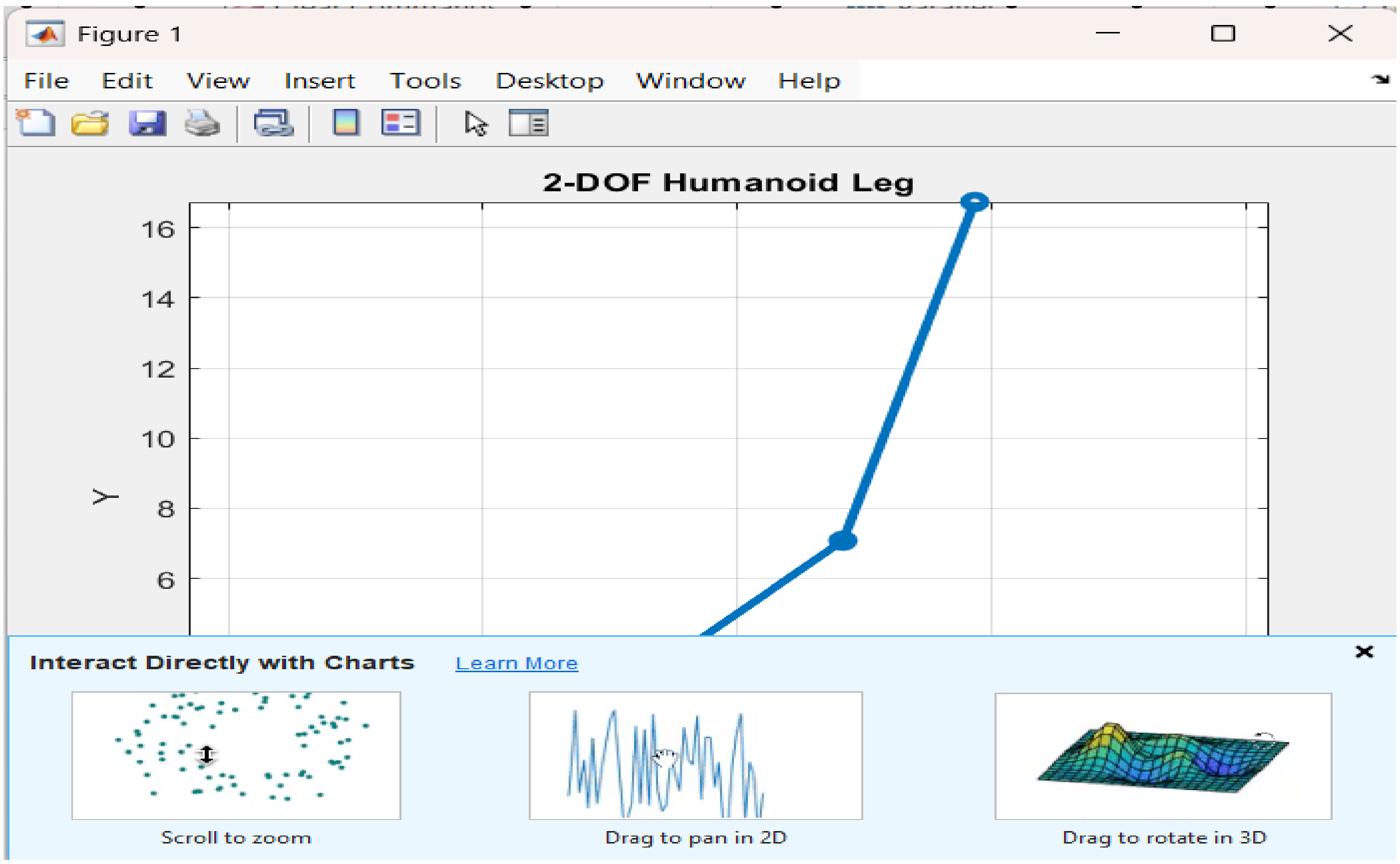


Figure 1
File Edit View Insert Tools Desktop Window Help
2-DOF Humanoid Leg
16
14
12
10
8
6
Y
Interact Directly with Charts
Learn More
Scroll to zoom
Drag to pan in 2D
Drag to rotate in 3D

Figure 1: Simulink model of the humanoid servo control system.

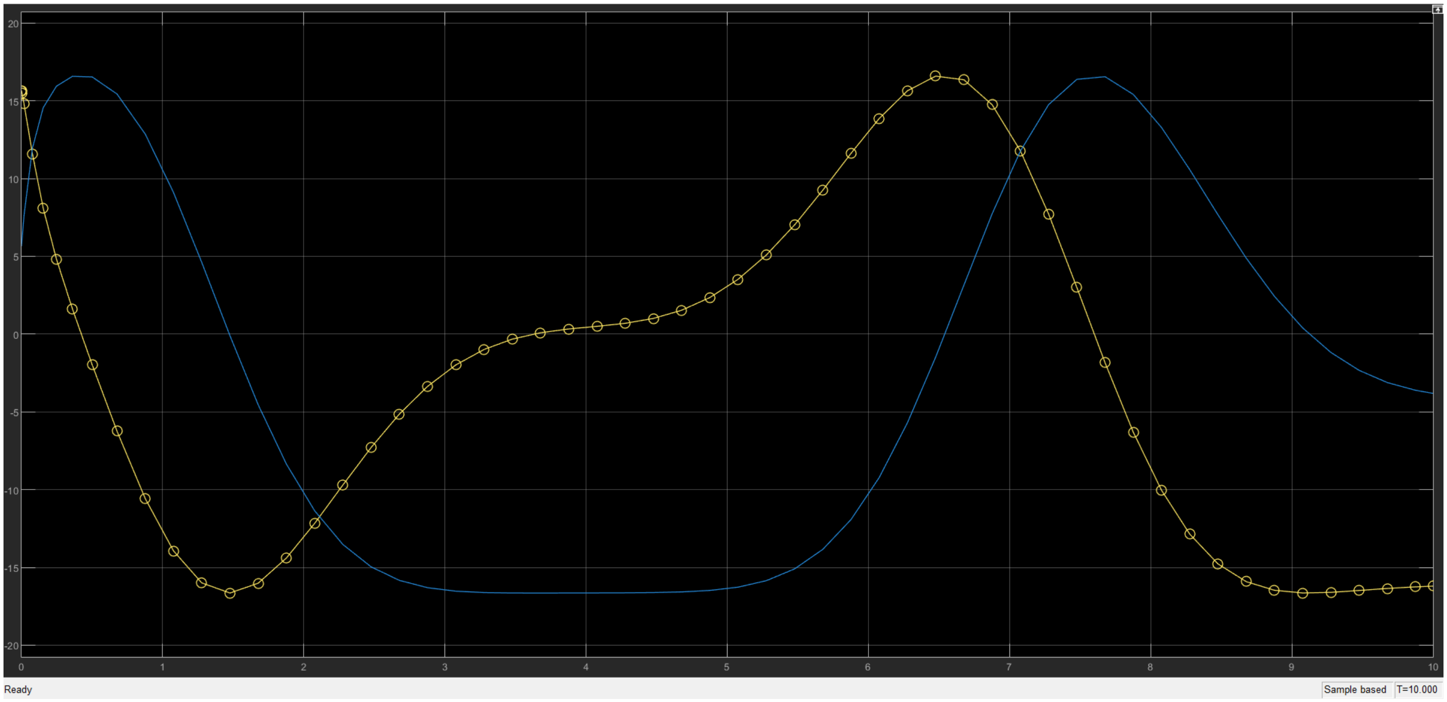


Figure 2: Servo motor dynamic response obtained from Simulink simulation.

Simulation Objectives

The simulation was conducted to:

- Validate servo motor dynamic response under load conditions
- Analyze motion smoothness during trajectory execution
- Evaluate overall system stability
- Verify safe operating limits of joint motion and torque
- Simulate the humanoid kicking motion prior to hardware implementation

1. Hardware Block Diagram

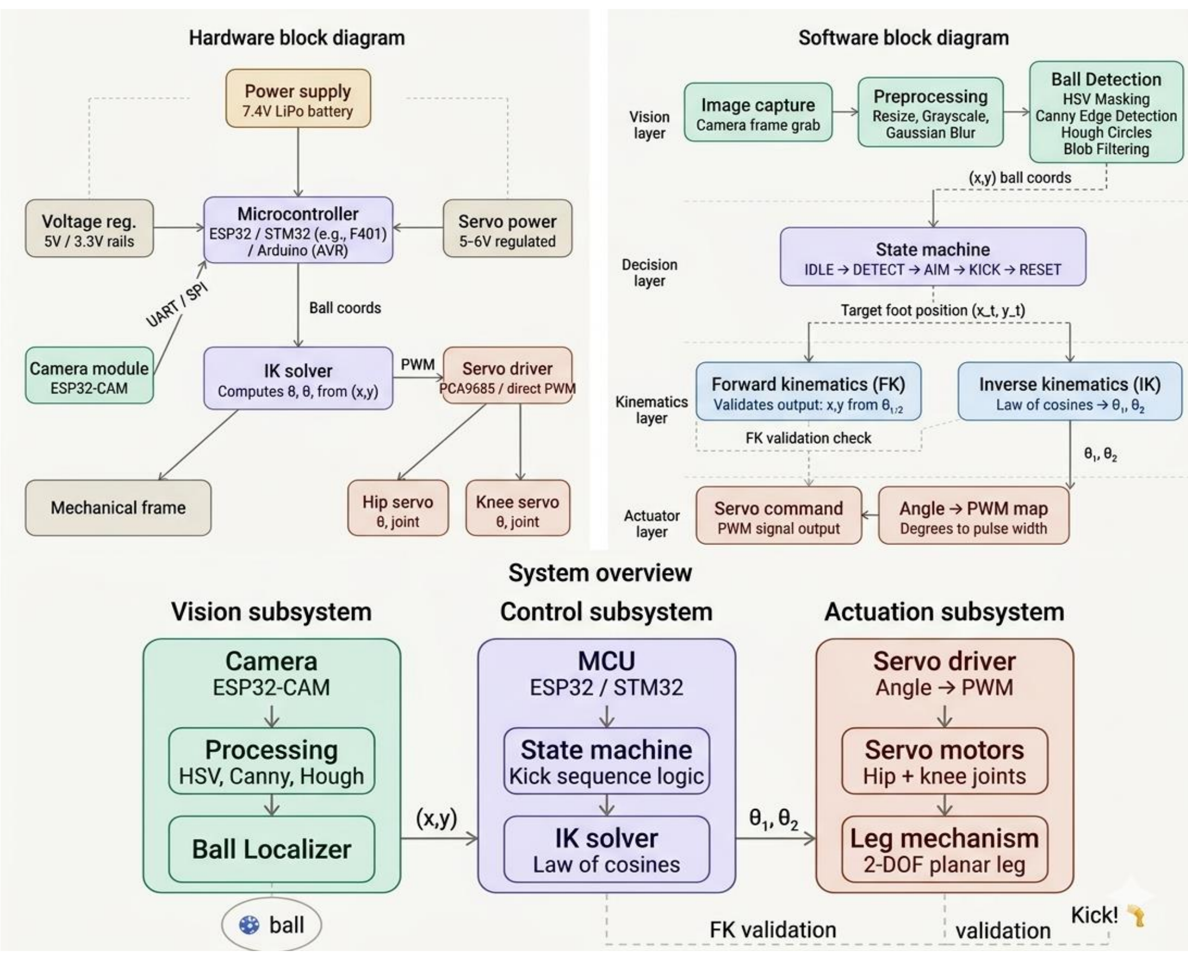
Illustrates the physical architecture of the system, showing the connections between the ESP32, ESP32-CAM, and servo motors, all powered by a regulated 10.8V LiPo battery. This diagram highlights the embedded hardware integration and power distribution across components.

2. Software Block Diagram

Represents the software pipeline and data flow starting from ball detection using the vision module, passing through the state machine for decision-making, and then moving to inverse/forward kinematics solvers for motion planning. The final output is PWM signals generated to control the servo motors.

3. System Overview

Provides a high-level abstraction of the complete robotic workflow, demonstrating how the system converts visual input from the camera into real-time mechanical actions, ultimately enabling the robot to detect the ball and execute a kicking motion.



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